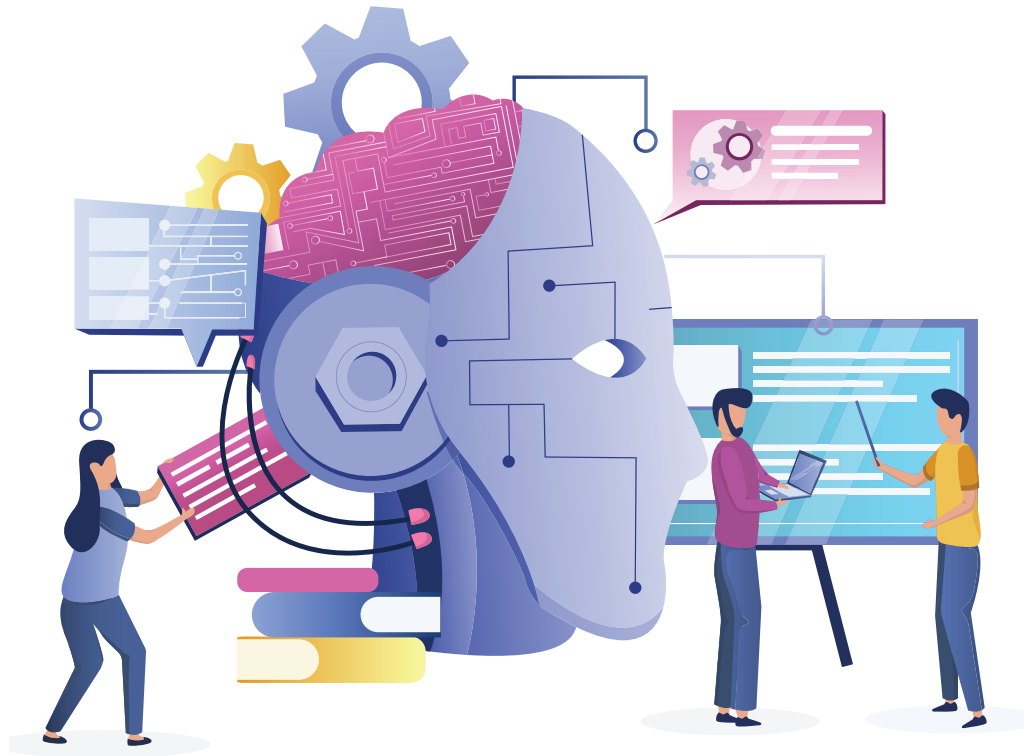


Python: Made for Making Machine Learning Models

Discover how Python, a language most programmers love, is also turning into the language of choice for developing machine learning models.



As we all know, machine learning (ML) automates decision-making while Python is a simple language with a clean syntax and numerous libraries. What's interesting is that Python has evolved with the times and is now the language of choice for machine learning. Before we find out why this is so, let's get acquainted with the basic programming components of Python used for machine learning.

Central to everything in Python are data structures — the lists, dictionaries, tuples, sets, etc, used to hold data in different forms. Lists are for ordered collections of items that can be iterated over and for mapping keys to the values; dictionary, an important aspect of labelled datasets; tuples provide immutability, useful in contexts where data integrity is critical; and sets deal with unique features — for instance, removing duplicates from a dataset.

Control structures include loops and conditional statements, and they allow logical flows and iterations over data. For example, you may need to loop through a dataset to clean missing values or apply a custom transformation

to each record. List comprehensions provide an elegant and direct Pythonic way to accomplish this, enhancing readability and reducing lines of code. Functions in Python allow forming the reusable logic encapsulated in them. For instance, whether we are talking about a function that normalises data or calculates accuracy, modular code with parameters and return values is what defines machine learning workflows. Also, higher-order functions in Python accept other functions as arguments, which is handy when filters or transformations are applied over datasets.

Working with files and external data is also fundamental in machine learning. Python's file handling capabilities, coupled with reading from CSV, JSON, and Excel files, help import data from different sources. With its advanced bulk functions for reading and manipulating tabular data, Pandas also makes this task simpler. Even though it is not strictly required for simple ML tasks, a knowledge of OOP or object-oriented programming is helpful in larger projects. The fact that classes are defined for models, datasets, or preprocessing pipelines leads to cleaner and more maintainable code.